SUBJECT: LIFE SCIENCES / HUMAN BIOLOGY

Molecular Genetics, DNA Replication & Cellular Pathways

Prepared for Offline AI Knowledge Ingestion & RAG Verification

1. Nucleic Acid Structure & Genetic Code

Deoxyribonucleic acid (DNA) is an antiparallel double helix composed of nucleotide monomers: adenine (A), thymine (T), guanine (G), and cytosine (C). Base pairing rules dictate that adenine pairs with thymine via two hydrogen bonds, while guanine pairs with cytosine via three hydrogen bonds.

2. The Central Dogma of Molecular Biology

- **Transcription:** RNA Polymerase II synthesizes messenger RNA (mRNA) from the template DNA strand in the 5' to 3' direction.
- **RNA Processing:** Pre-mRNA undergoes 5' capping (7-methylguanosine), 3' polyadenylation (poly-A tail), and spliceosomal excision of non-coding introns.
- **Translation:** Ribosomal complexes read triplet codons on the mature mRNA to assemble polypeptide chains using aminoacyl-tRNA molecules.

3. CRISPR-Cas9 Gene Editing

Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR) combined with the Cas9 endonuclease enables targeted double-strand breaks guided by synthetic single-guide RNA (sgRNA). Repair occurs via non-homologous end joining (NHEJ) or homology-directed repair (HDR).